$3 = 11 = 2^1 + 2^0$

43210

bit i $\quad 2^i$

---

xor $\quad 5 \oplus 6 =$ 101 | 0 = 3

110 | 0

011

Or $\quad 5 | 6 =$ 101 | 0

110 | 0

111 | 0

And $\quad 5 \& 6 =$ 101

110

100

---

n

$7 = 111$

$x < n \quad$ n: 1 0

x: 0 1

n: 0111

x: 0

0 1 1 1

0 $\quad 2^2$

$+ 0 \cdot 2^2 + 2 \cdot 2^1 = 2$

$+ 1 \cdot 2^1 + 1 = 3$

$+ 2 \cdot 2 + 0 = 2$

$+ 3$

$2^i > 2^0 + \dots + 2^{i-1}$

$1 << 0 = 1$

$1 << 1 = 2$

010

$11 << 1 = 110$

Int : 32 bits

Long Long : 64 bits

	2	1	0	
0	0	0	0	2^i
1	0	0	1	
2	0	1	0	
3	0	1	1	
4	1	0	0	
5	1	0	1	
6	1	1	0	
7	1	1	1	

$$\sum_{i=0}^{50} \frac{n}{2^{i+1}} + n \% (2^{i+1}) - 2^i + 1$$

$\oplus$ $\quad 5 \oplus 5 = 0$

$+ \; - \quad \oplus \to \oplus \quad r:$

$f(1, i) -$

$f(l, r) = f(r) \oplus f(l-1)$

$x \oplus x = 0$

$c(i) = 1, \dots, i$

cantidad de a_i con bit j encendido

$c(l, r) = c(r) - c(l-1)$

$c(r) \mp c(l-1)$ (2)

1, 3, 2

bit 0 : 1, 1, 0

$\underbrace{1, 1}_{0}$

$\underbrace{1\ 1\ 1}_{1}$

bit i $\quad 2^i$

$\underbrace{l \quad r}$

impar bit encendidos

$C(r)$ impar
cont. $C(l)$ par

$C(r)$ par
cont. $C(l)$ impar

$f(l, r) = l \oplus (l+1) \oplus (l+2) \ldots \oplus (r)$

$f(l, r) = f(r) \oplus (l-1)$

$f(r) =$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

0, 1, 3, 0, 4, 1, 7, 0, 8, 1, 11, 0

$$f(r) = \begin{cases} 0 & r \equiv 3 \ (4) \\ 1 & r \equiv 1 \ (4) \\ r+1 & r \equiv 2 \ (4) \\ r & r \equiv 0 \ (4) \end{cases}$$

0 ... r

dp con bitmasks

$\{3,5,7\} \to 111$

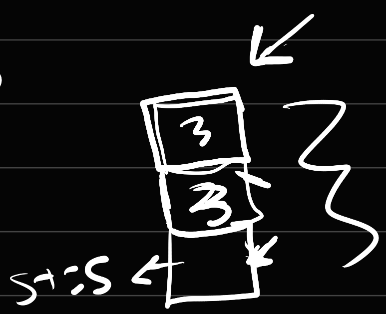
$\{5,7\} \to 011$

$\{7\} \to 001$

$|S| \leq 64$ LL

Guard Mark

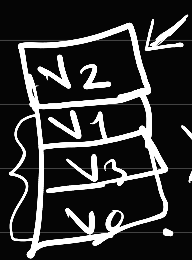
n vacas



w, st, h

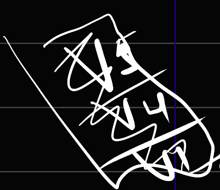
max V

V2: h, st, w



$x + h$

$V = \min(st, V - w)$



$S = \{V1, V3, V0\} = 01011$

$dp[S] = \max V$

$|S| = 20$

2^{20}

$S \to$

$$dp[S \mid (1 << i)] = \max \min(s[i],\ dp[S] - w[i])$$

$$O(2^n \cdot n)$$

$n = 2$ $\qquad$ $2^2 = 4$

0: 00

1: 01

2: 10

3: <u>11</u>